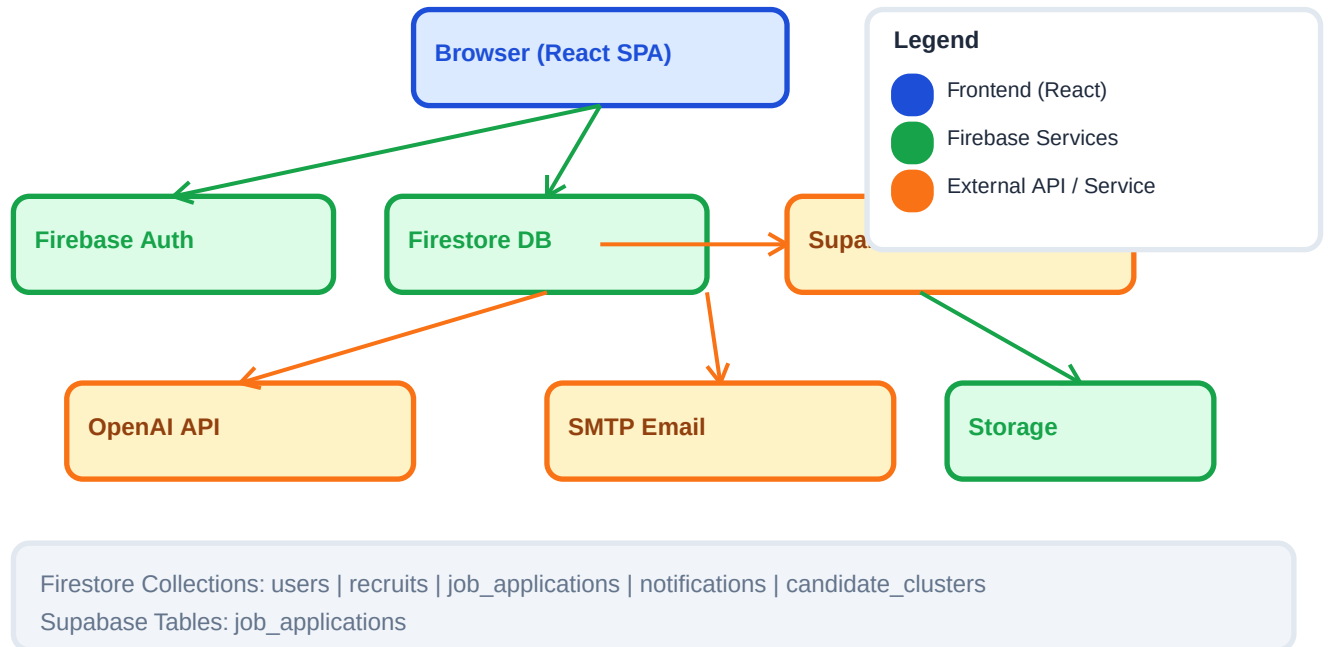


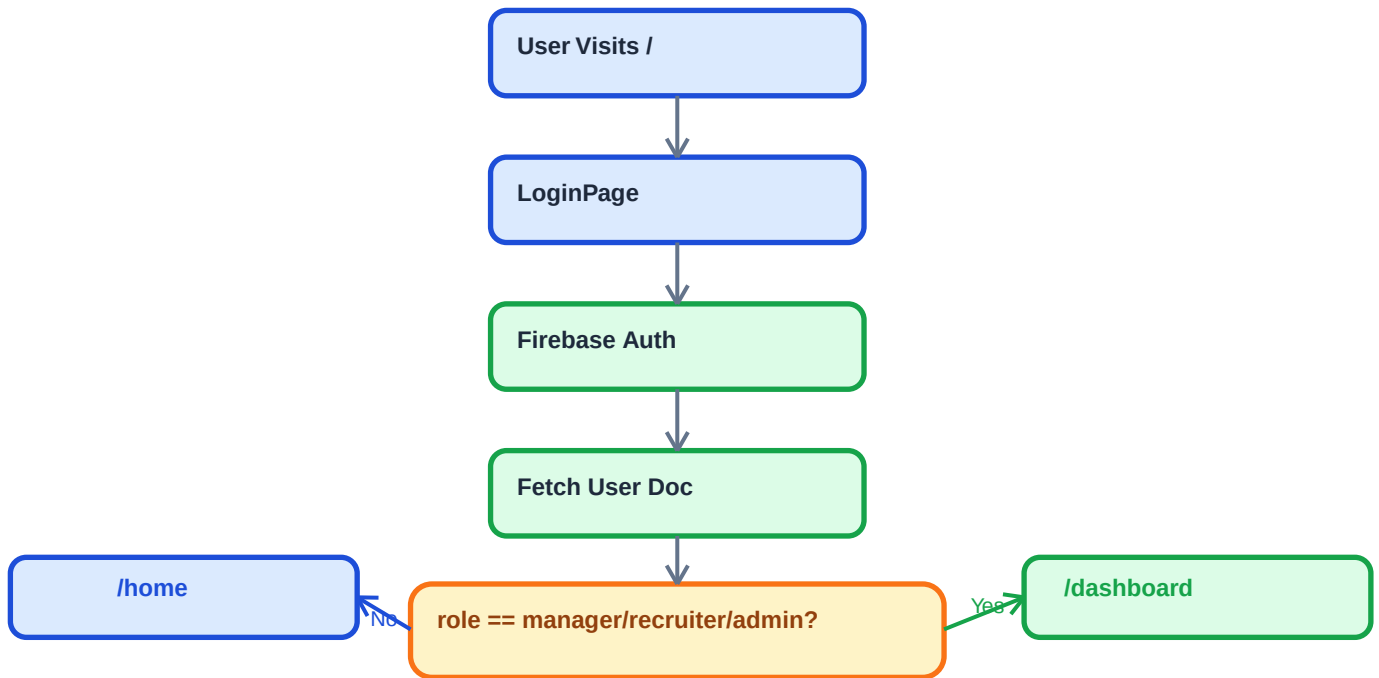
Key Data Flows

Indian Infra Jobs — Architecture & Flow Diagrams

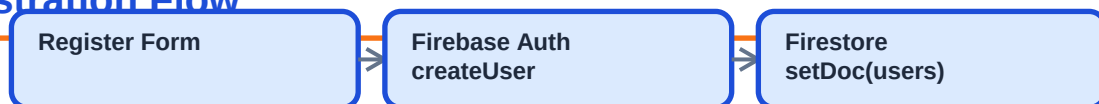
1. High-Level System Architecture



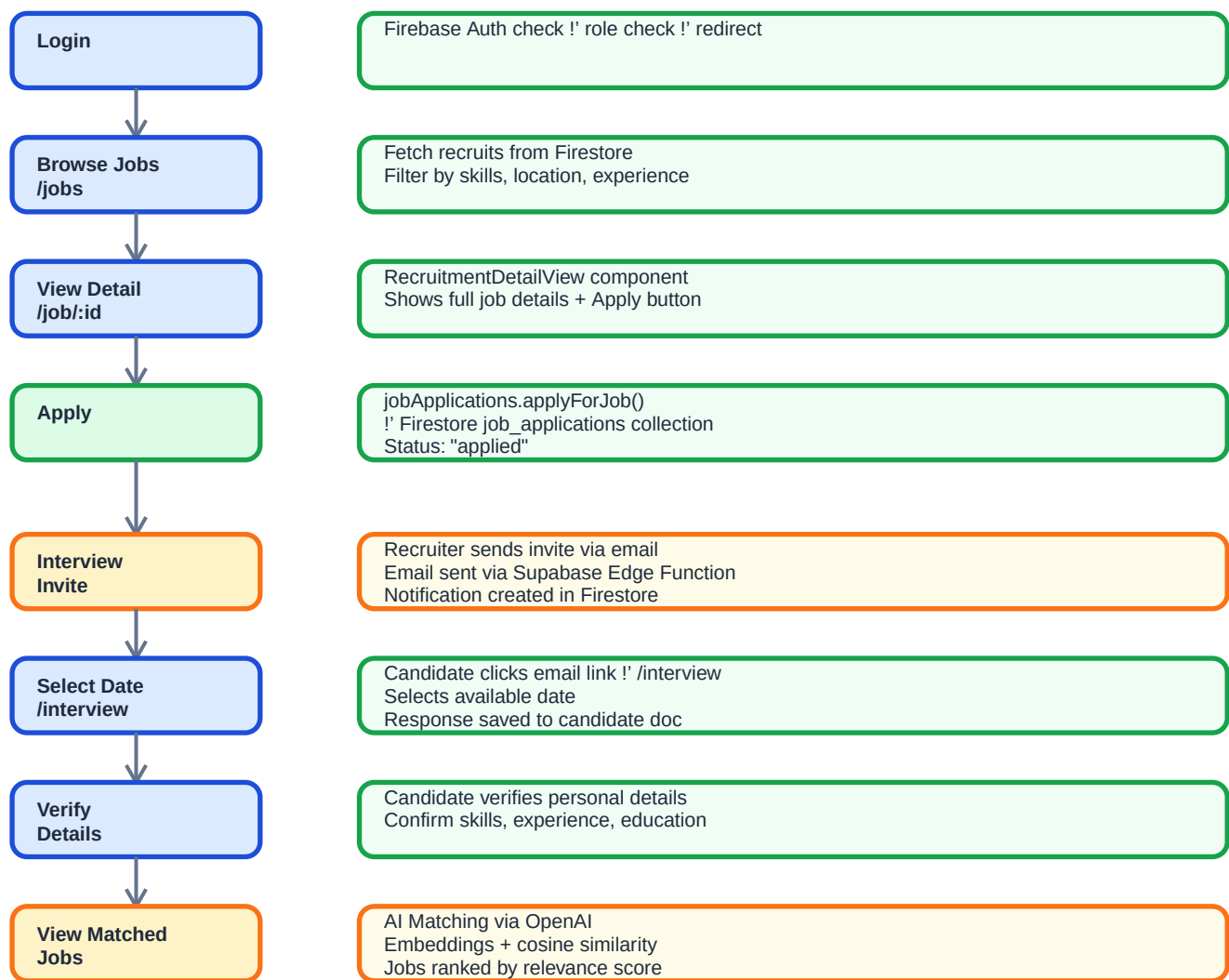
2. Authentication & Role-Based Routing



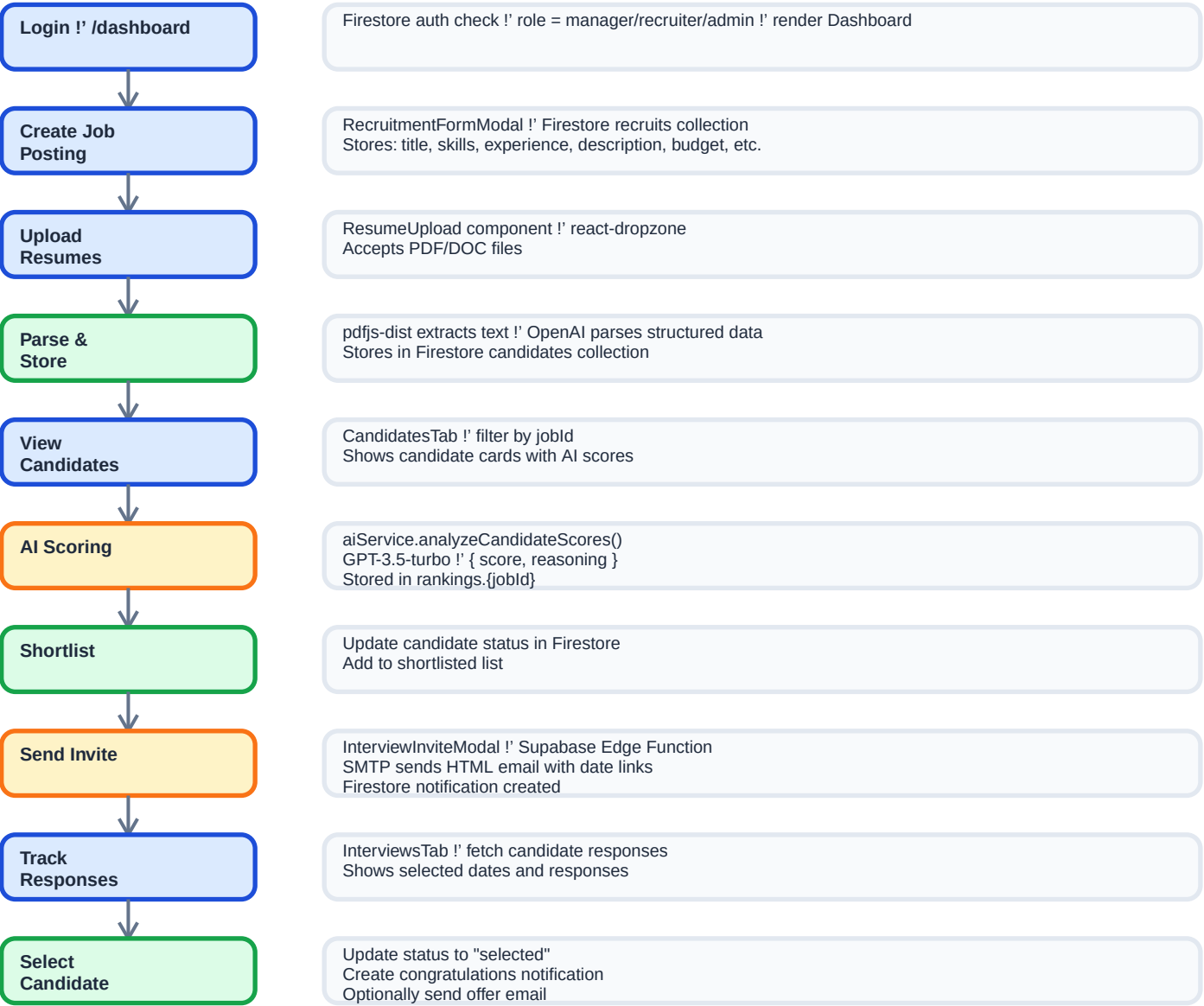
Registration Flow



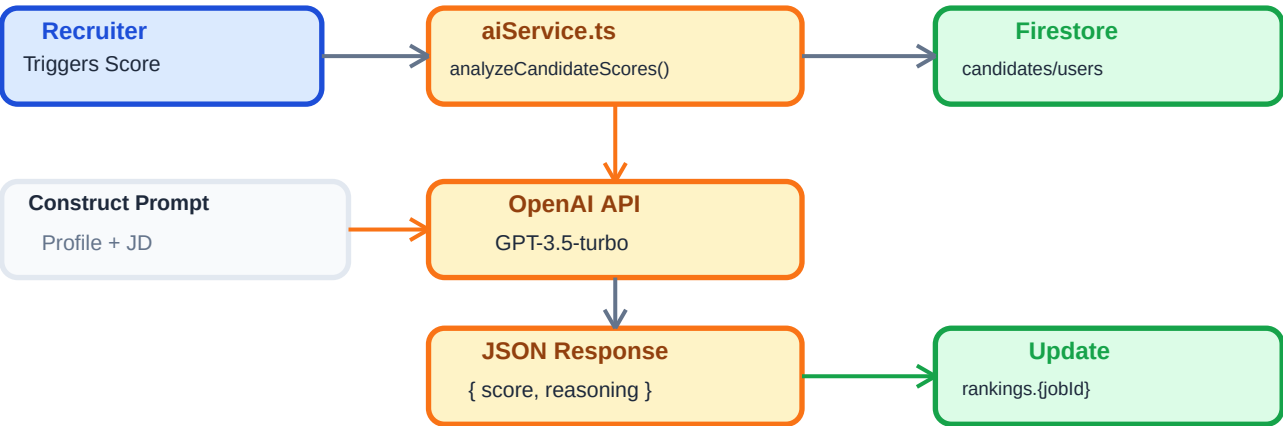
3. Job Seeker — End-to-End Flow



4. Recruiter — End-to-End Flow



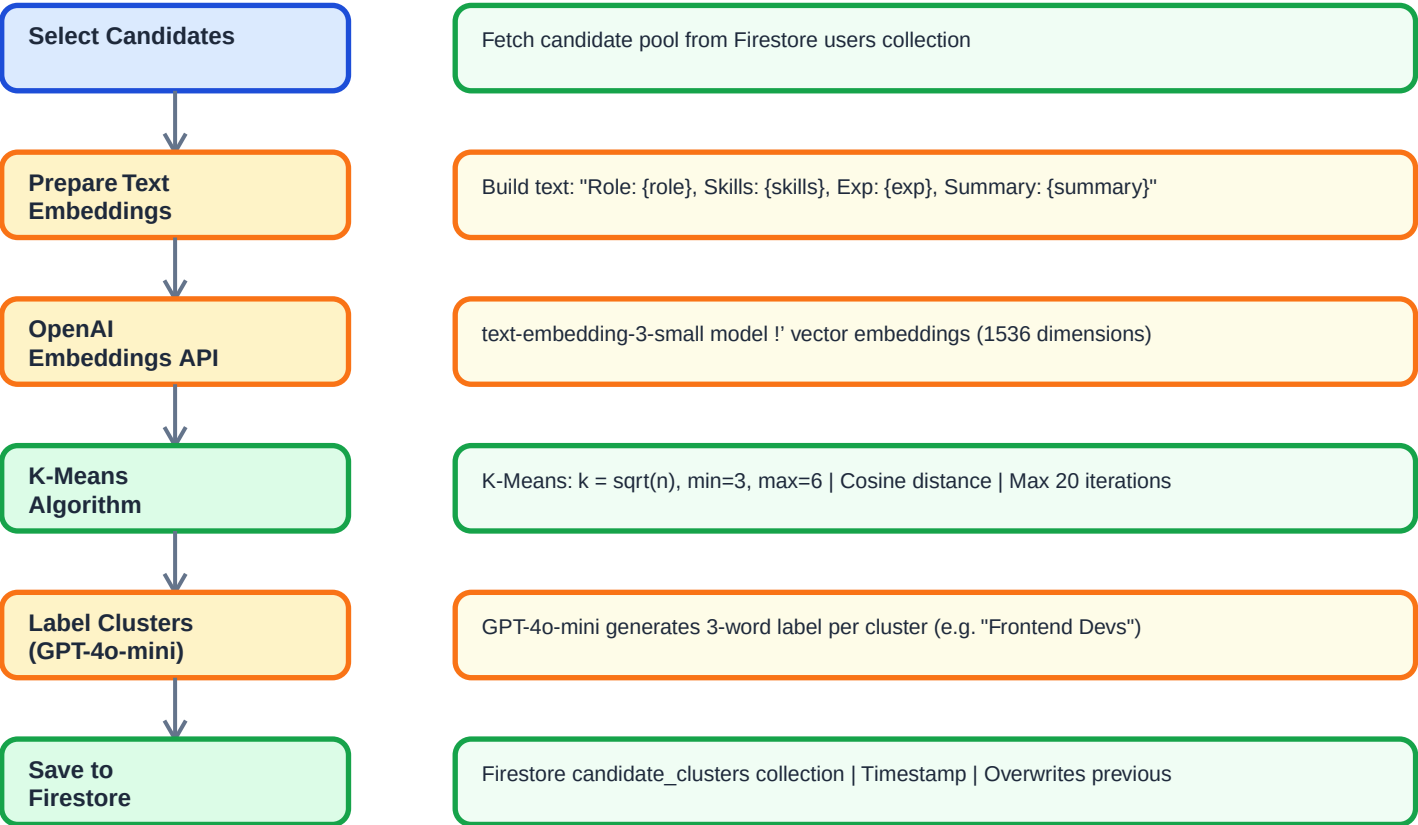
5. AI Scoring — Detailed Data Flow



Prompt Template:

"You are an expert technical recruiter. Analyze the following candidate against the Job Description.
JOB TITLE: {jobTitle} | JOB DESCRIPTION: {jdText} | CANDIDATE PROFILE: {candidateProfile}
Provide a JSON response with: score (0-100), reasoning (1 sentence)."
Model: gpt-3.5-turbo | Response format: JSON object | Cost: ~\$0.002 per candidate

6. Candidate Clustering — Architecture



K-Means Algorithm Detail:

1. Initialize k centroids from first k vectors
 2. Assign each point to nearest centroid (cosine distance)
 3. Recompute centroids as cluster mean | 4. Repeat until stable (max 20 iterations)
- Cosine Distance = $1 - (A \cdot B / (||A|| \times ||B||))$

7. Interview Invitation — Email Flow



Email Content:

- Role & round type
- Available interview dates (listed)
- Interview panel names
- "Confirm availability" action link
- "Decline" link
- Action URL: {baseUrl}/interview?candidateId={id}

Candidate Response Flow:



8. Notification System & Summary

Notification Types:		
interview_invite	Recruiter sends invite	Candidate
verify_details	Interview scheduled	Candidate
congratulations	Candidate selected	Candidate
manager_invite	Admin adds member	New Manager
premium_request	Recruiter requests premium	Admin
premium_approved	Admin approves premium	Recruiter
premium_rejected	Admin rejects premium	Recruiter

Summary

The Indian Infra Jobs platform follows a modern React SPA architecture with:

- Firebase for authentication, primary database (Firestore), file storage, and hosting
- Supabase for secondary data storage and edge functions (email sending)
- OpenAI API for AI-powered candidate scoring, clustering, and job matching
- Role-based routing: Users see /home, Managers/Admins see /dashboard
- Real-time notifications via Firestore + email via SMTP edge functions

Generated: 8/7/2026

Indian Infra Jobs — Key Data Flows Architecture